

Nayoung Kim

UX Designer

Email: nayoungkim0826@gmail.com

Mobile: +82 10 7728 5366

<https://08k26ny.github.io/nayoung/>

ABOUT ME

I am a UX/HRI designer who translates complex robotic technologies into user-centered interaction experiences. My work focuses on robot service planning, interaction prototyping, and usability evaluation for AI and robotic systems. I aim to bridge the gap between technical possibilities and real user needs through research-driven design.

EDUCATION

- | | |
|------------------------|---|
| Sep 2021 –
Aug 2023 | Yonsei University , Seoul, Korea, Republic of <ul style="list-style-type: none">• Master of Science in Industrial Engineering• Major: Human-Computer Interaction (HCI), Advisor: Yong Gu Ji• GPA: 4.06 / 4.3 |
| Mar 2017 –
Aug 2021 | Sungshin Women University , Seoul, Korea, Republic of <ul style="list-style-type: none">• Bachelor of Social Science• Major: Psychology• GPA: 4.11 / 4.5 |

WORK EXPERIENCE

- | | |
|------------------------|---|
| Sep 2024 –
Present | UX Lead, AI Robotics Lab, Sogang University , Seoul, Korea, Republic of
(Project 1) Elderly-friendly Shared Autonomy for Teleoperated Robots in Superaged Society
(Project 2) Development of a continual intelligence-enhancing RaaS framework based on human-robot shared control <ul style="list-style-type: none">• Designed LLM-based robot teleoperation interfaces and conducted usability testing for AI robotic systems• Designed end-to-end interaction flows from early concepts to high-fidelity prototypes• Collaborated with engineers to align technical development with user-centered design principles• Conducted psychology-based human-robot interaction (HRI) research and translated findings into UX requirements |
| Sep 2023 –
Feb 2024 | UX Designer, AI Robotics Institute, Korea Institute of Science and Technology (KIST)
(Project) Advanced Technology Development Project for Efficient Installation and Operation of Quarantine Treatment Facilities <ul style="list-style-type: none">• Planned AI robotic control services for quarantine treatment facilities and conducted scenario-based usability testing• Designed psychology-based AI robotic research as part of individual studies and presented findings at IROS |

Project

Aug 2024 –
Present

Elderly-friendly Shared Autonomy for Teleoperated Robots in Superaged Society

: Research project funded by the National Research Foundation of Korea (NRF)

- Planned robot service scenarios through task analysis and field observations
- Analyzed user characteristics and developed HRI design guidelines through desk research
- Designed wireframes and robot interaction prototypes using Figma and ProtoPie
- Identified user needs and pain points through user research and usability testing
- Collaborated with social workers, developers, and engineers to translate UX insights into robot system and interface requirements
- Submitted research findings to an SCI-indexed international journal (under review)

Aug 2023 –
Feb 2024

Investigating Behavioral and Cognitive Changes Induced by Autonomous Delivery Robots in Incidentally Copresent Persons

: Individual research funded by Korea Institute of Science and Technology (KIST) project of Intelligence on Multi-Robot Autonomy

- Planned and validated UX scenarios for autonomous delivery robots
- Investigated user perceptions of delivery robot behaviors in shared public spaces
- Presented research findings at IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

Aug 2021 –
Feb 2022

Interface Optimization Research for Autonomous Driving UX

: Research project funded by Hyundai Motor Company

- Conducted user research to identify pain points and needs in existing infotainment systems
- Designed multimodal infotainment interfaces, including navigation, to enhance UX
- Performed usability testing and data analysis to validate the enhanced system

PUBLICATIONS

Nov 2025

Kim, N. Y., ... & Nam, C. J. (2026). Hey, Robot! Don't Cut in Line: Designing Queuing Behaviors of Delivery Robots with Multiple Stakeholders. *IEEE Robotics and Automation Letters*

Aug 2023

Shin, HR., **Kim, NY.**, ... & Ji, Y. G. (2024). Evaluating and eliciting design requirements for an improved user experience in live-streaming commerce interfaces. *Computers in Human Behavior*, 150, 107990

Apr 2026

Ko, JH., Jung, HC., **Kim, NY.** & Nam, CJ. DEX-Mouse: A Low-cost Portable and Universal Human Interface with Force Feedback for Data Collection of Dexterous Robotic Hands. *IEEE Robotics & Automation Letters* (Under Review)

Apr 2026

Kim, NY., ... & Lee, SO. Beyond Assistance: Self-Motivated Engagement of Older Adults with Robots in Everyday Life. *International Journal of Human-Computer Interaction* (Under Review)

Nov 2025

Lee, JH., ..., **Kim, NY.** & Nam, CJ. HEART: Coordination of Heterogeneous Expert Agents for Physically Grounded Robotic Task Planning. *IEEE Robotics & Automation Letters* (Under Review)

Nov 2025

Kim, NY., ... & Nam, CJ. Mitigating Barriers to Workforce Engagement Associated with

Aging: Designing Situation-Aware Interfaces for Teleoperated Robots., International Journal of Social Robotics (Under Review)

Sep 2025 Lee, SB., ... **Kim, NY.** & Nam, CJ. Human Factor-Aware Adaptive Task Allocation for Multi-Human Multi-Robot Remote Supervision. IEEE Transactions on Human-Machine Systems (Under Review)

CONFERENCE PRESENTATIONS

2024 **Kim, N.,** & Kwak, S. S. (2024, October). Investigating Behavioral and Cognitive Changes Induced by Autonomous Delivery Robots in Incidentally Copresent Persons. In 2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) (pp. 2514-2519). IEEE.

2023 **Kim, N.,** ... & Kim, S. Analysis of User Experience Requirements for Continuous Use of Digital Healthcare Services for Female Infertility. *PROCEEDINGS OF HCI KOREA 2023*, 402-407

2022 **Kim, N.,** Kim, G., Ji, Y.G., Differences in User Information Perception Depending on the Delivering Modality of the Voice Assistant's Search Results in a Driving Situation., *Proceedings of the ESK Conference 2022*, 284.

PATENT

Dec 2024 **Kim, N.,** Kim, S. "A Medication Management Device Capable of Detecting Emergencies." Korea Patent (10-2747601-0000)

AWARDS

Jun 2022 **Grand Award Smart Pillbox: Elderly Emergency Detection Pillbox for Sustainable Society**
: Smart City Platform Idea Awards, Yonsei University
• Grand Award, Team Work
